IPv6 Networking Assignment Report

Sample documentation and code for the requested IPv6 tasks.

Task 1: Static IPv6 Address Configuration

Linux example: sudo ip -6 addr add 2001:db8:1::100/64 dev eth0. Verify using ip -6 addr show or ifconfig. Windows example: netsh interface ipv6 add address "Ethernet" 2001:db8:1::100 and verify with ipconfig.

Task 2: Unique Local Address (ULA)

Assign fd00:1234:abcd:1::100/64 to the interface. Test another host such as fd00:1234:abcd:1::101 using ping -6 fd00:1234:abcd:1::101. Successful replies confirm connectivity.

Task 3: Temporary IPv6 Address

Enable privacy extensions and observe temporary IPv6 addresses. After reboot or disconnect/reconnect, the temporary address may change while the stable address remains available. Document the observed behavior in a text file.

Task 4: Zomato-style Search Script

Python script included below. It stores restaurant names and IPv6 addresses and searches by partial name.

Task 5: IPv6 Subnetting Scenario

Subnet A: 2001:db8:100:1::/64, Subnet B: 2001:db8:100:2::/64, Subnet C: 2001:db8:100:3::/64. Example host address assigned from Subnet B: 2001:db8:100:2::50/64.

Sample Python Script:

```
restaurants = [
{"name":"Pizza Hub","ipv6":"2001:db8:100:1::10"},
{"name":"Burger Town","ipv6":"2001:db8:100:2::20"},
{"name":"Spice Garden","ipv6":"2001:db8:100:3::30"}
]
query = input("Enter restaurant name: ").lower()
for r in restaurants:
if query in r["name"].lower():
print(r["name"], r["ipv6"])
```